

Syllabus and Assessment

EART22001 Environmental Modelling

Syllabus

Academic year: 2026–2027. Semester 1; 10 credits.

Activity	Time and location
Lectures	Mondays, 11:00–12:00; Simon Building, Theatre D.
Practicals	Thursdays, 13:00–15:00; Humanities Bridgeford Street, rooms 2.1 and 2.2.

Programme information

Core module for:

- Environmental Science (PEP);
- Environmental Science (Atmospheric and Climate Science);
- Environmental Science (Ecology, Evolution and Conservation).

Optional for:

- Earth & Planetary Science (Geochemistry);
- Earth & Planetary Science (Planetary).

Instructors

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Assessment

Assessment	Weighting
Online test 1 - available from 30 October 2026	0%
Online test 2 - 30 November to 4 December 2026	50%
Modelling essay - due 18 December 2026	50%

Computing and course material

Practical work will normally use the course Linux server. The server has a **course-specific username and password**, supplied on Canvas; these are **not** your central University credentials. The *Tools of the Trade* handout gives Windows and macOS instructions for SSH, file transfer and the optional Docker workflow.

[Module webpage](#) Canvas is the authoritative location for assessment links, server login details, announcements and any updated instructions.

Attendance, practice and feedback

Attendance at the timetabled teaching and practical activities is expected. You will receive feedback by discussing work with the instructors and graduate teaching assistants (GTAs), by checking practical results and interpretation in class, and through discussion of common issues arising from assessments and modelling exercises.

The questions in the *Pressure, Temperature, Humidity, Visibility and Particulate* and *Dynamical Models* chapters are an important part of your preparation. Work through them yourself before looking at the worked answers; use the answers to check your reasoning and bring anything you do not understand to class.

Online tests

Canvas is the authoritative source if an assessment announcement or instruction is subsequently updated.

Online test 1 - formative/practice (0%)

Available from 30 October 2026.

- This test is designed to help you check your understanding and does not contribute to the final module mark.
- Questions will focus on the models used in the course, the assumptions behind them, and the interpretation of practical results.
- Use it diagnostically: if a question exposes something you do not understand, return to the relevant practical, work through the reasoning, and ask about it before the summative test.
- The test will be accessed through Canvas. Any closing date or updated instructions will be stated there.

Online test 2 - summative (50%)

30 November to 4 December 2026.

- The format is similar to the practice test, but this assessment contributes 50% of the module mark.
- Questions may draw on all material covered before the test, including the reasoning and interpretation developed in the practicals.
- Do not revise only definitions or commands. Make sure you can explain what a model is doing, identify its assumptions, interpret output, and reason about how changing an input or parameter should affect the result.
- Check Canvas for the exact opening and closing times and any final instructions.

A useful way to prepare. Work through the questions in the scientific chapters without looking at the worked answers first. Then revisit each practical and ask yourself: What question was the model answering? What were the important inputs and assumptions? What did the output mean physically? Which result would change if an assumption or parameter changed? If you can answer those questions without simply following the practical instructions, you are preparing at the right level.

Modelling essay - summative (50%)

Due: 18 December 2026. Target length: **4 pages** (approximately 1500–2000 words); a maximum of 5 pages may be used if necessary. You may use up to **2 figures**. Submit through Canvas/Turnitin; check Canvas for the exact submission time and final assessment brief.

What are you being asked to do?

The brief is to *discuss the use of an environmental modelling approach and its assumptions*. You may write as an environmental consultant explaining a modelling approach to a manager or decision-maker, or as a scientist discussing it in the style of a research paper.

The important word is **discuss**. Do not stop at describing what the model does. Explain *why* the approach is useful, what theory or empirical relationships it contains, what has been simplified or neglected, and how those choices affect what can be concluded from the results.

Plan the argument before you write

Choose a modelling approach used in the module and keep the focus tight. You may compare a small number of complementary models if the comparison helps you make a clear point about assumptions, complexity or appropriate use. Before drafting, try to complete the sentence: “*This essay will explain/evaluate ... because ...*”. If that sentence is vague, narrow the topic.

Part	Questions worth answering
Context	What environmental problem is being addressed? Why is a model useful here? What does it predict or explain?
Model	What are the essential equations, relationships or empirical ideas? What goes in and what comes out? Include only the mathematics needed to understand the approach.
Assumptions	What is simplified, parameterised, fixed or neglected? Why might those assumptions be reasonable, and when might they fail?
Evaluation	What are the strengths and limitations? When would you trust the result, and when would you be cautious? How could observations test or constrain the model?
Conclusion	Answer the question you set up at the start. State what the approach is useful for and what a user must remember when interpreting it.

What stronger essays tend to do

- maintain a clear argument rather than presenting disconnected facts;
- distinguish between **what the model calculates** and **what the real environmental system does**;
- connect assumptions to consequences - explain what an assumption makes possible and what behaviour it may prevent the model from representing;
- interpret evidence rather than simply report it, and use technical detail only where it supports the argument;
- evaluate the modelling approach in the conclusion rather than merely repeat the introduction.

Literature, figures and equations

Use additional literature where it strengthens the discussion. A few well-chosen sources that you understand and use critically are better than a long list of references. Cite sources for important assumptions, methods, data and claims. Figures should earn their space: refer to each one in the text and use the caption to explain what the reader should learn from it. Equations are useful when they clarify the model, but the essay should not become a derivation exercise.

Before submitting: check that you answered the brief, explained the important assumptions, discussed every figure/equation, supported important claims with sources, and checked the final PDF, references and page limit.

Indicative essay marking descriptors

These descriptors indicate what distinguishes stronger work. The formal University criteria and the Canvas assessment brief take precedence if they differ.

Grade	Description
81% or more	Outstanding work. Excellent command of the modelling approach and its assumptions; clear, accurate and independent critical judgement; a well-structured argument; effective use of evidence; professional presentation; figures/captions and citations are accurate and informative.
70%–80%	First class. Strong command of the material and underlying principles, with clear critical evaluation of the modelling choices and limitations. Well organised and clearly written, with appropriate use of sources and figures.
60%–69%	Upper second class. Good knowledge of the model and its scientific context, with some effective evaluation of assumptions and limitations. Mostly clear structure and presentation, with appropriate evidence and only minor errors or omissions.
50%–59%	Lower second class. Generally correct knowledge and a reasonable summary of the modelling approach, but analysis may be more descriptive than critical or may omit important assumptions/limitations. Presentation is adequate but may contain some errors or weakly integrated evidence.
40%–49%	Third class. Some relevant knowledge is present, but the explanation or conclusions are incomplete, unclear or only weakly connected to the principles of the model. Limited critical evaluation and/or significant presentation or referencing problems.
1%–39%	Fail. Essential material is absent or seriously misunderstood; the discussion is substantially flawed, unsupported or too unclear to demonstrate the required understanding.
0%	Normally reserved for non-submission, subject to the University's current assessment and academic-malpractice regulations.